

Cao Wu

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Education

Politecnico di Milano

Milan, Italy

M.Sc. IN HIGH PERFORMANCE COMPUTING ENGINEERING

Sep. 2024 - Dec. 2026

- Core coursework: Parallel Computing, Partial Differential Equations, Advanced Computer Architecture, Machine Learning, and Distributed Systems.

Sichuan University

Chengdu, China

B.ENG. IN COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2018 - Jun. 2022

Work Experience

Bilibili

Shanghai, China

GOLANG BACKEND SOFTWARE DEVELOPMENT ENGINEER (FULL-TIME)

Jul. 2022 - Aug. 2024

- Built core settlement services for Bilibili's live-streaming business with Go, MySQL, Redis, Kafka, Hive, and ClickHouse.
- Promoted within the first year and became a main developer of the anchor settlement system.
- Supported settlement flows with cumulative processed rewards exceeding RMB 1.3 billion from Jul. 2022 to Aug. 2024.

Projects

GrowYumi

Personal WeChat Mini Program

FOUNDER

Feb. 2022 - Present

- Built and operated a personal WeChat Mini Program for couples, reaching **100,000+** registered users.
- With access to the WeChat advertising platform, monthly income can reach RMB 1,000 during peak periods.

A Simplified vLLM Core

LLM Inference Systems

PERSONAL PROJECT

Dec. 2025 - Present

- Rebuilt core vLLM-style inference components from scratch in PyTorch/CUDA, including paged KV-cache block management, continuous batching, and a lightweight LLM serving engine.
- Implemented a FlashAttention-style prefill CUDA kernel and benchmarked it against dense baselines, achieving about 10% lower prefill latency than a naive CUDA attention kernel (4.64 ms vs. 5.15 ms).
- Designed KV-cache writeback and INT8 KV-cache quantization, reducing KV memory usage by about 50% (0.0078 GB to 0.0039 GB) with low quantization overhead (0.052 ms quantize, 0.145 ms dequantize).

RGB-Infrared Fusion for Vehicle-Scene Segmentation

Sichuan University

GRADUATION PROJECT

Sep. 2020 - Nov. 2021

- Improved semantic segmentation under low-light and glare conditions with RGB-infrared fusion.
- Used PyTorch and a temporal-difference method to improve recognition efficiency.

Selected Awards

- 2021 **Member of ACM Algorithm Training Team (One of 32 members)**, Sichuan University
- Apr. 2021 **Second Prize, Southwest Division (Top 15%)**, China University Computer Contest
- Oct. 2021 **Provincial Second Prize (Top 20%)**, Lanqiao Cup C/C++ Algorithm Programming Competition
- Oct. 2020 **Silver Medal (Top 10%)**, China Collegiate Algorithm Design & Programming Challenge Contest

Skills

Languages & Frameworks

Python, C++, CUDA, C, Go, PyTorch, Hugging Face

LLM Inference

KV cache, PagedAttention, continuous batching, vLLM, Transformers, Qwen

CUDA Kernels

Matrix multiplication, transpose, reduction, FlashAttention-style prefill, GPU memory optimization